Jyothi Swaroop Ganapavarapu

AI Engineer | RAG Systems · Agentic AI · LLM Fine-tuning

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Summary

AI Engineer and MS Data Science candidate (GPA 3.9, UNT, May 2026) with **2+ years** of experience designing and deploying Generative AI systems — specializing in RAG pipelines, agentic AI, LLM fine-tuning, and MCP protocol engineering. Published researcher in *Science of the Total Environment* (Elsevier, Impact Factor 8.2). Shipped production-grade AI systems using LangGraph, LangChain, Groq, and HuggingFace — spanning hybrid retrieval, multi-agent benchmarking, knowledge distillation, and tool-server integration — all on free-tier APIs.

Skills

- **AI / LLM Frameworks:** LangChain, LangGraph, HuggingFace PEFT, Groq, LlamaIndex
- **RAG & Retrieval:** FAISS, ChromaDB, BM25, BGE-large, GraphRAG, RRF Fusion, Cross-encoder Reranking
- **Agentic AI:** ReAct, Plan-Execute, Tool Calling, Multi-Agent Orchestration, LangGraph State Machines, MCP
- **LLM Fine-tuning:** QLoRA, LoRA, 4-bit NF4 Quantization, Qwen3-4B, Llama 3.1, Mistral, Gemini 2.5 Flash
- **Evaluation & Benchmarking:** RAGAS, ROUGE-L, BLEU, Tool-Call Accuracy, Latency Profiling
- **Prompt Engineering:** Chain-of-Thought, Few-Shot, RAG Prompting, System Prompt Optimization
- **Deployment & MLOps:** FastAPI, Streamlit, HuggingFace Spaces, GitHub, Docker, Selenium
- **Languages & Data:** Python, SQL, Pandas, NumPy, Scikit-learn, Plotly

Projects

•Enterprise Agentic RAG Platform — *Live Demo | GitHub*

LangGraph · LangChain · Groq · ChromaDB · BGE-large · BM25 · GraphRAG · Tavily · FastAPI · Streamlit

- Engineered a production-grade RAG platform using a **20-node LangGraph state machine** with 4-signal hybrid retrieval — dense BGE-large vectors, BM25 sparse search, GraphRAG entity traversal, and temporal scoring — fused via **Reciprocal Rank Fusion (RRF)** with cross-encoder reranking across **481 indexed document chunks**.
- Constructed a **183-entity, 177-relationship knowledge graph** via LLM extraction enabling graph-augmented retrieval; recording **BM25 fire rate 90%** and avg 2.8/4 retrieval signals per query, validating multi-signal robustness.
- Evaluated across **18 RAGAS metrics on 10 benchmark queries** — achieved **answer relevancy 1.00, 0% hallucination rate, 100% response success rate**, avg 3.6 citations/response, and 17/20 pipeline nodes executing per query. Deployed live on HuggingFace Spaces using **100% free-tier APIs**.

•Multimodal Agentic Assistant — ReAct vs Plan-Execute Benchmark — *Live Demo | GitHub*

LangGraph · LangChain · Groq (Llama 3.3-70B / 8B / Mixtral) · Tavily · E2B · FastAPI · Streamlit · Plotly

- Architected a Multimodal Agentic Assistant implementing two agent paradigms — **ReAct** (Thought→Act→Observe loop) and **Plan-Execute** (Planner→Executor→Synthesizer nodes) — benchmarked head-to-head across **50 queries spanning 5 task categories**.
- Integrated **5 heterogeneous tools** — Tavily web search, E2B sandboxed Python execution, Groq LLaMA Vision (image analysis), AST-based calculator, and Wikipedia search — enabling true multimodal capability across text, code, and image modalities.
- Developed a custom benchmark harness computing ROUGE-L, tool-call accuracy, and latency across 3 Groq models; **ReAct achieved 87.3% tool-call accuracy vs 71.2% for Plan-Execute** with **34% lower average latency**. Launched Streamlit dashboard with Plotly analytics.

•Low-Cost Resume Optimization via LLM Distillation into a Fine-Tuned SLM — *GitHub*

Qwen3-4B · QLoRA (r=16, 4-bit NF4) · Gemini 2.5 Flash · HuggingFace PEFT · Selenium · FastAPI · Python

- Designed an end-to-end **LLM knowledge distillation pipeline** — scraped **249 job descriptions** and extracted **1,800+ resumes** as domain corpus; used **Gemini 2.5 Flash Batch API as teacher model** to generate **1,530 high-quality (instruction, output) training pairs**.
- Fine-tuned **Qwen3-4B using QLoRA** (rank=16, alpha=32, 4-bit NF4 quantization), reducing VRAM footprint from **28GB to under 8GB** while maintaining task quality; delivering **90% cost reduction vs GPT-4** inference with **<120ms latency** on domain-specific resume optimization tasks.
- Validated distillation pipeline on held-out pairs demonstrating GPT-4-level output quality from a **4B-parameter student model**; confirmed full pipeline from data collection through inference via live FastAPI frontend demo.

•MCP-Powered AI Research Assistant — *Live Demo | GitHub*

fastmcp · LangGraph · LangChain · Groq (LLaMA 3.3-70B) · arXiv API · Tavily · Streamlit

– Architected a **custom MCP (Model Context Protocol) server** using `fastmcp` exposing **5 standardized research tool endpoints** — `search_arxiv`, `web_search`, `summarize_paper`, `format_citation`, and `compare_papers` — with `format_citation` at **<5ms** and `search_arxiv` at **sub-800ms** latency.
– Engineered a **LangGraph ReAct agent** as the MCP client connecting to the server at runtime via the MCP protocol; agent dynamically chains tools across multi-step research pipelines (search → summarize → compare → cite) averaging **9 tool calls per query** at **36.5s end-to-end latency**; powered by Groq LLaMA 3.3-70B delivering structured 5-section comparisons and 3–5 sentence summaries.
– Implemented **persistent multi-turn memory** with JSON-backed session storage enabling cross-session research continuity; agent generates APA and BibTeX citations on demand. Shipped live on **HuggingFace Spaces** using **100% free-tier APIs** (Groq, Tavily, arXiv).

Experience

•**Graduate Student Assistant** *Feb 2025 – May 2026*
University of North Texas Denton, TX
– Developed Power BI and Excel dashboards visualizing student grade distributions and department-level academic performance metrics, enabling faculty and department staff to identify performance trends across courses.
– Processed and analyzed student grade data using Excel, generating structured performance reports to support department-level academic decision-making.
– Collaborated with faculty to organize and maintain academic datasets, contributing to research documentation and departmental reporting workflows.

•**Research Assistant** *Mar 2023 – May 2024*
KL University Vijayawada, India
– Conducted field data collection at **3 coastal sites** (Veraval, Porbandar, Kelwa) on the North-West Coast of India; gathered ground-truth seaweed biomass measurements to validate satellite-derived spectral indices and support peer-reviewed research published in *Science of the Total Environment* (Elsevier, IF 8.2).
– Processed and analyzed **Landsat-8 OLI satellite imagery** — band selection, DN to TOA reflectance conversion, PCA-based dimensionality reduction, pan sharpening, dark object subtraction — applying NDVI, FAI, and SEI indices to detect and map seaweed cover changes across 5 years of data (2018–2022).
– Developed **Python-based ML pipelines** using scikit-learn (regression, classification, clustering) on multi-year satellite datasets; benchmarked SEI against NDVI and FAI across 3 coastal sites, demonstrating SEI superiority in seaweed identification accuracy.

•**AI/ML Virtual Intern** *Mar 2022 – Jun 2022*
All India Council for Technical Education (AICTE) · AWS Academy Program India · Remote
– Completed a structured AI/ML program through AICTE EduSkills covering supervised/unsupervised learning, model evaluation, and deep learning fundamentals.
– Constructed end-to-end ML pipelines across data collection, preprocessing, feature engineering, model training, and evaluation on real-world datasets.
– Applied Python and scikit-learn to solve classification, regression, and clustering problems using industry-standard workflows.

Publications

•**A Novel Seaweed Detection Image Processing and Validation Framework: A Pragmatic Study on Natural Seaweed Beds along North-West Coast of India** *May 2025*
– *Science of the Total Environment*, Elsevier · Vol. 978 · **Impact Factor: 8.2** · 10.1016/j.scitotenv.2025.179296
– Proposed and validated a novel remote sensing framework integrating Landsat-8 satellite imagery with **advanced image processing techniques** (PCA, spectral band enhancement, atmospheric correction) — compared NDVI, FAI, and SEI detection performance across Veraval, Porbandar, and Kelwa with in-situ biomass validation; **SEI demonstrated superior accuracy** over NDVI and FAI for seaweed habitat identification.

Education

•**University of North Texas** *Aug 2024 – May 2026*
Master of Science in Data Science ***GPA: 3.9/4.0*** Denton, Texas
– **TGS R.B. Toulouse Merit Scholarship** (2024) — $1,000 merit award; qualified for **in-state tuition waiver** under Texas state policy, resulting in **40% tuition reduction** for the MS program.

•**KL University** *Jul 2020 – May 2024*
B.Tech in ECE — Specialization: Data Science & Big Data Analytics ***CGPA: 9.53/10*** Vijayawada, India

Certifications

- **Microsoft Certified: Azure AI Fundamentals** · Microsoft · Sep 2022
- **Microsoft Certified: Azure Data Fundamentals** · Microsoft · Oct 2022
- **Microsoft Certified: Azure Fundamentals** · Microsoft · Sep 2022
- **AWS Academy Graduate — Machine Learning Foundations** · Amazon Web Services · Apr 2022